

1. Title, goal statement and conventions

Project title: Ad-Vocates

Goal: Create a reliable and efficient tool that improves browsing performance while giving users greater privacy, security, and control over their web experience.

Conventions: Solid boxes represent components built by the team, circles represent external dependencies the team does not build, arrows for the D0 diagram represent interfaces, arrows for the data-flow diagram represent data with its form labeled

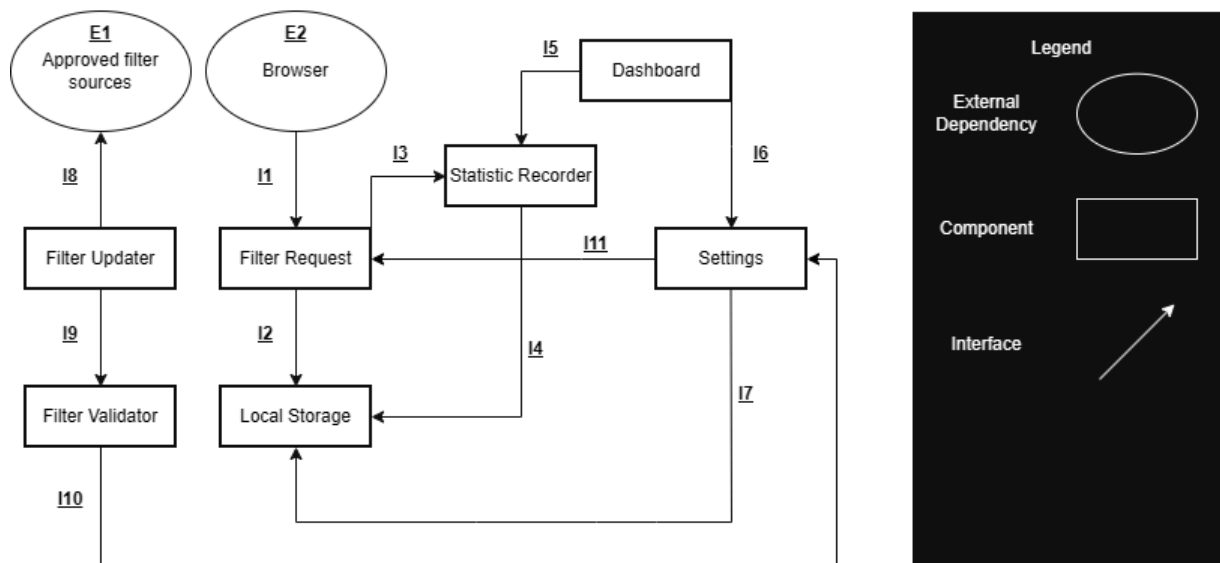
Basic input: Browser request data, user protection settings and filter rules from approved sources

Basic output: Block/allow output, and dashboard showing blocked requests and estimated data saved

2. Block diagram (the D0 diagram)

Ad-Vocates

Goal: Create a reliable and efficient tool that improves browsing performance while giving users greater privacy, security, and control over their web experience.



3. Component responsibility table

Component	Interface in/out	Primary owner	User story
Dashboard - present controls, settings and statistics to user	In : None Out: I5, I6	Max Marsh	US-02, 03, 05
Settings - manage filters and protection settings	In: I6, I10 Out: I7, I11	Max Marsh	US-03, 04, 05
Statistic Recorder - produce diagnostic stats from blocking event data	In: I3, I5 Out: I4	Dylan Shackleford	US-01, 02
Filter Request - block or allow browser request based on filters/settings	In: I1, I11 Out: I2, I3	Max Marsh	US-01, 03, 05
Local Storage - persistent app records stored on user machine	In: I2, I4, I7 Out: None	Max Marsh	US-02, 03, 04, 05
Filter Updater - fetch updated/current filter data from approved sources	In: None Out: I8, I9	Dylan Shackleford	US-04
Filter Validator - validate safety of retrieved filter set	In: I9 Out: I0	Dylan Shackleford	US-04

4. Interface specification table

ID	Named and typed in/out	Data format	Protocol	Error behavior / handler
I1 Browser to Filter Request	In : requestId: string, requestUrl: string, resourceType: string, pageHost: string Out: result:	Structured object	Browser extension request	Filter Request allows request if filtering fails and tries to log error in Local Storage

	enum(allowed, blocked)			
I2 Filter Request to Local Storage	In: action: enum(readFilters, logError), errorMsg: string (opt) Out: filters: string[], status: enum(success, failed)	JSON	Local read/write	Filter Request keeps current filters if read fails, if log fails filtering continues
I3 Filter Request to Statistic Recorder	In: eventId : string, category : enum(ad, tracker, redirect, other), estimatedBytes: int Out: None	JSON	Local internal event	Statistic Recorder ignores duplicate events and continues if recording fails
I4 Statistic Recorder to Local Storage	In: action: enum(read, write), blockedTotal : int (opt), estimatedBytes: int (opt) Out: blockedTotal: int, estimatedBytes: int, status: enum(success, failed)	JSON	Local read/write	Statistic Recorder keeps current total in memory if save fails and reports unavailable statistics if read fails
I5 Dashboard to Statistic Recorder	In: query: enum(getTotal) Out: blockedTotal : int, estimatedBytes: int, status: enum(success, failed)	JSON	Local query/response	Dashboard shows “Statistics currently unavailable” if query fails

I6 Dashboard to Settings	In: action: enum(read,write), enabled: bool (opt), allowlist: string[] (opt) Out: enabled: bool, allowlist: string[], status: enum(success, failed)	JSON	Local request/response	Settings rejects invalid addresses, keeps prior config, and returns error to Dashboard
I7 Settings to Local Storage	In: action: enum(read,write), enabled: bool (opt), allowlist: string[] (opt), filters: string[] (opt) Out: enabled: bool, allowlist: string[], filters: string[], status: enum(success, failed)	JSON	Local read/write	Settings keeps prior config if save fails and reports failure through Dashboard
I8 Filter Updater to Approved Sources	In: sourceUrl: string Out: filterText: string, statusCode: int	Plain text filter	HTTPS request	Filter Updater retrieves a set amount of hours later if download fails and keeps existing filters
I9 Filter Updater to Filter Validator	In: sourceUrl: string, filterText: string Out: valid: bool, errorMsg: string (opt)	Structured object	Local validation request/response	Filter Validator rejects unapproved sources or invalid filters, existing filters stay active
I10 Filter Validator to Settings	In: filters: string[] Out: status: enum(success, failed)	JSON	Local request/response	Settings keeps previous filters if new validated filters cannot be saved/applied
I11 Settings to	In: enabled:	JSON	Local	Filter Request

Filter Request	bool, allowlist: string[], filters: string[] Out: status: enum(success,fai led)		request/response	keeps previous config if new config fails, Settings reports error to Dashboard
----------------	--	--	------------------	---

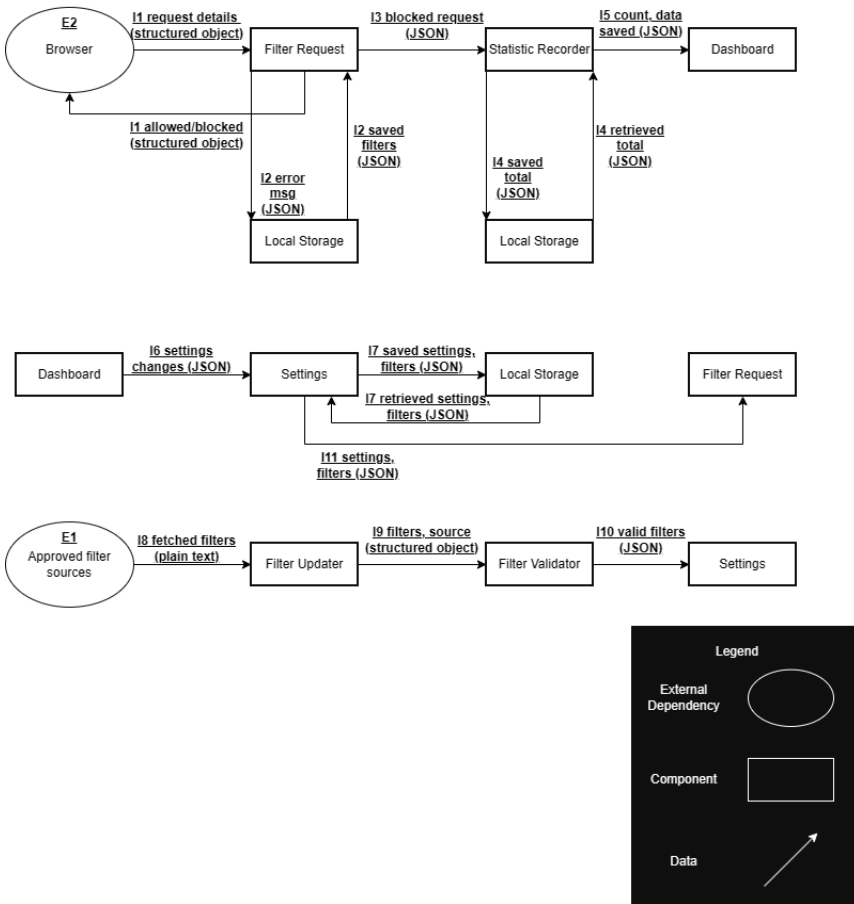
An example payload for I3 would be the following:

```
{
  "eventId": "request1",
  "category": "ad",
  "estimatedBytes": 1234
}
```

5. Data-flow diagram

Ad-Vocates

Goal: Create a reliable and efficient tool that improves browsing performance while giving users greater privacy, security, and control over their web experience.



Requests are blocked before they finish loading.

6. Architecture pattern and justification

7. Decision log